



University of Asia Pacific
Department of Computer Science and Engineering
CSE 450: VLSI Lab

LAB REPORT

Experiment Number: 04

Experiment Title: Design and Simulation of an 8-Bit Register Using NAND Gates and Flip-Flops

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1 Title of the Experiment

Design and Simulation of an 8-Bit Register Using NAND Gates and Flip-Flops

2 Objectives

The objectives of this experiment are:

1. To design an 8-bit register using structural VHDL.
2. To construct the register using flip-flop modules.
3. To implement the required flip-flop logic using NAND gates.
4. To understand hierarchical and structural design in VHDL.
5. To create and analyze the RTL schematic of the 8-bit register using Xilinx ISE.
6. To simulate the designed register and verify its functionality using a testbench.
7. To understand how individual flip-flops can be combined to construct a multi-bit storage element.

3 Software and Tools Used

The following tools were used for this experiment:

- Xilinx ISE 14.7
- VHDL
- ISim Simulator
- Computer system for design and simulation

4 Design Methodology

The 8-bit register was designed using a hierarchical structural approach. Instead of directly describing an 8-bit register using behavioral VHDL, the design was divided into smaller modules. The basic hierarchy is:

NAND Gate → Flip-Flop → 4-Bit Register → 8-Bit Register

The complete 8-bit register consists of two 4-bit register sections:

- **REG_LOW** - stores the lower 4 bits
- **REG_HIGH** - stores the upper 4 bits

Each 4-bit register consists of four flip-flops, while the flip-flop is constructed using NAND gates.

5 RTL-Schematic Diagrams

5.1 Top-Level RTL Schematic

The top-level RTL schematic represents the complete 8-bit register and shows how the lower and upper 4-bit registers are connected. The 8-bit register is divided into `REG_LOW` and `REG_HIGH`.

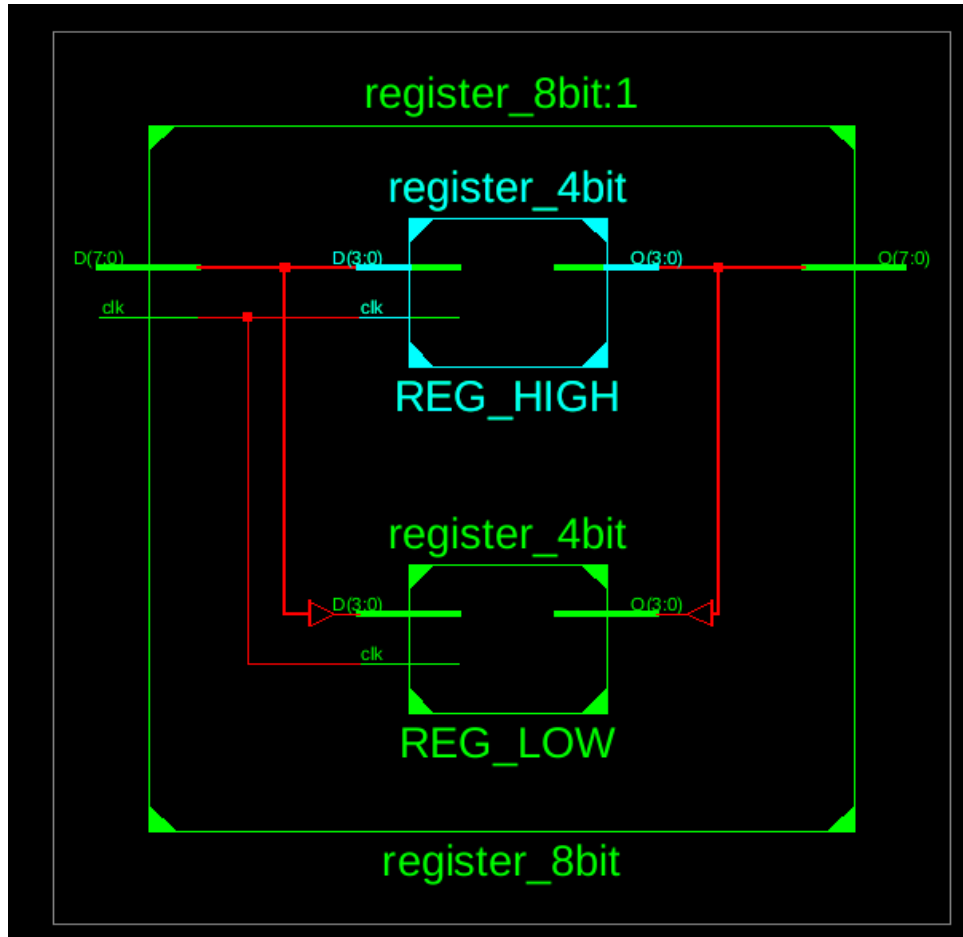


Figure 1: Top-Level RTL Schematic of the 8-bit register.

5.2 4-Bit Register RTL Schematic

Each 4-bit register is constructed from four flip-flops. The same structural design is used for both `REG_LOW` and `REG_HIGH`.

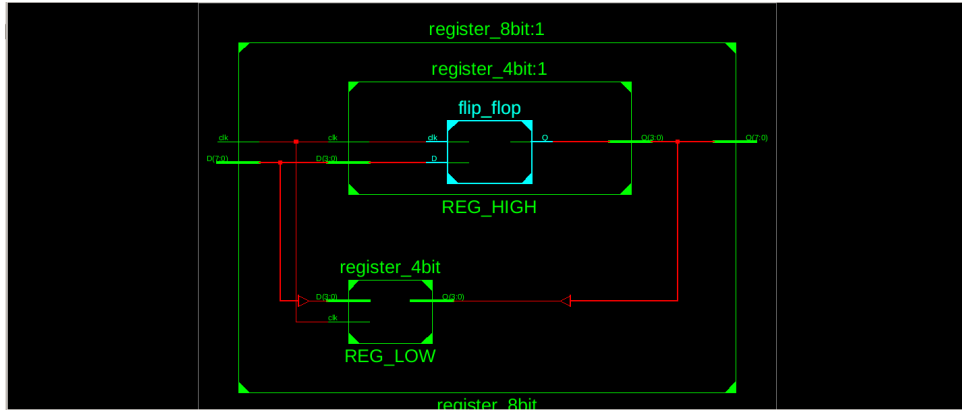


Figure 2: RTL Schematic of the 4-bit register.

6 Flip-Flop Design Using NAND Gates

The individual flip-flops were implemented using NAND gates. The purpose of using NAND gates at this level is to demonstrate the construction of sequential logic from basic logic gates rather than using a pre-built register component.

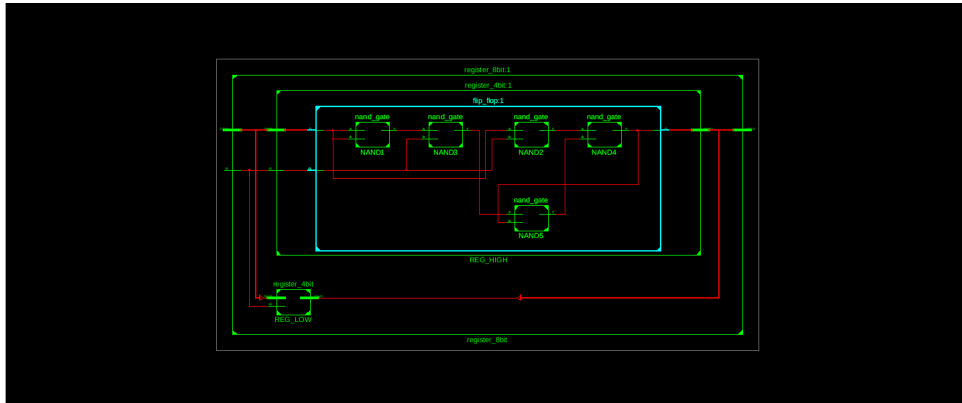


Figure 3: Internal NAND-gate implementation of a flip-flop.

7 NAND Gate Implementation

The NAND gate is used as the fundamental logic component in the flip-flop design. Multiple NAND gates are interconnected to construct the flip-flop.

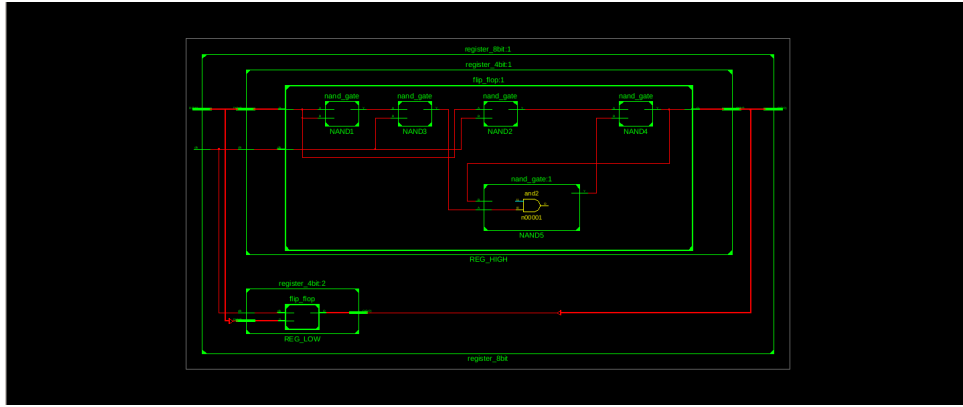


Figure 4: NAND gate used as the basic building block.

8 VHDL Hierarchy

The complete design was implemented using structural VHDL. The design contains separate VHDL modules for:

1. `nand_gate.vhd`
2. `flip_flop.vhd`
3. `register_4bit.vhd`
4. `register_8bit.vhd`
5. `register_8bit_tb.vhd`

This modular approach makes the design easier to understand, simulate, test, and modify.

9 GitHub Repository

All VHDL source codes, testbenches, and related project files have been uploaded to a GitHub repository.

GitHub Repository Link:

<https://github.com/IstiaaNabil/Vlsi-Lab-Report>

10 Simulation

The designed 8-bit register was simulated in Xilinx ISE ISim using a VHDL testbench. The testbench provides input signals to the register and observes the resulting output to verify that it accepts 8-bit input data, stores the data correctly, produces the corresponding outputs, and that the hierarchical design behaves as expected.

10.1 Simulation Waveform

The simulation waveform demonstrates different input combinations being applied to the register.

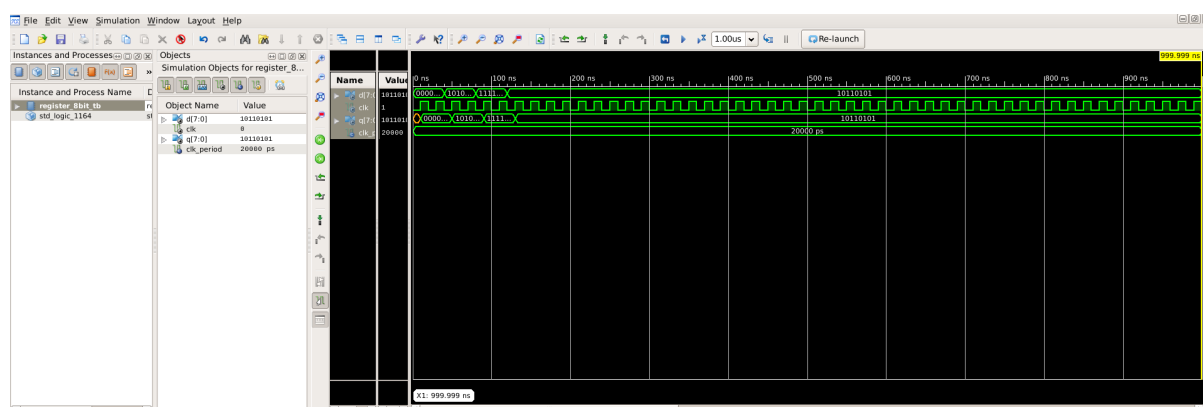


Figure 5: ISim simulation waveform of the 8-bit register.

11 Simulation Analysis

The simulation waveform verifies the functional behavior of the complete 8-bit register. When valid 8-bit input data is applied to the data input lines and the required clock transition occurs, the flip-flops store the corresponding binary values. For example, when the input transitions to $D[7:0] = 10101010$, the register successfully stores and produces the output $Q[7:0] = 10101010$. Subsequent inputs of 11110000 and 10110101 are similarly stored and reflected on the output lines, confirming the correct data-storage behavior of the hierarchical structural design.

12 Conclusion

In this experiment, an 8-bit register was designed and simulated using structural VHDL in Xilinx ISE. The register was constructed hierarchically from smaller digital components. The fundamental building block was the NAND gate, which was used to implement the flip-flop. Multiple flip-flops were then combined to form 4-bit registers, and two 4-bit registers were combined to create the complete 8-bit register. The RTL schematic and Xilinx ISE hierarchy demonstrated the internal structure of the design, while the simulation waveform was used to verify the functional behavior of the register. The experiment provided practical understanding of hierarchical structural VHDL design, NAND-based sequential logic, flip-flop construction, register organization, RTL synthesis, and digital circuit simulation.